

SOLVED PRACTICE WORKBOOK

Arithmetic-and-Commercial-Math - CSAT Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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Identity: csat-03: learner-v2:g1 | Approval: false

BASIC MCQS / REMEDIATION

Diagnostic set - Questions 1-16

Q1. Practice variant 1: What is 15% of 200?

A. 30 B. 31 C. 29 D. 32

Q2. Practice variant 1: A value of 1000 rises by 10% and then falls by 5%. What is the final value?

A. 1046 B. 1045 C. 1044 D. 1047

Q3. Practice variant 1: Groups of 20 and 30 have averages 40 and 60. What is their combined average, rounded to the nearest integer?

A. 53 B. 51 C. 52 D. 54

Q4. Practice variant 1: An article costs Rs 500 and is sold at 20% profit. What is the selling price?

A. 601 B. 599 C. 602 D. 600

Q5. Practice variant 1: Find the simple interest on Rs 1000 at 5% per annum for 2 years.

A. 100 B. 101 C. 99 D. 102

Q6. Practice variant 1: Find the compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually.

A. 211 B. 210 C. 209 D. 212

Q7. Practice variant 1: Rs 360 is divided in the ratio 2:3. What is the first share?

A. 145 B. 143 C. 144 D. 146

Q8. Practice variant 1: A invests Rs 1000 for 12 months and B invests Rs 1500 for 8 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?

A. 501 B. 499 C. 502 D. 500

Q9. Practice variant 2: What is 20% of 220?

A. 44 B. 45 C. 43 D. 46

Q10. Practice variant 2: A value of 1000 rises by 11% and then falls by 6%. What is the final value?

A. 1044 B. 1043 C. 1042 D. 1045

Q11. Practice variant 2: Groups of 21 and 31 have averages 41 and 61. What is their combined average, rounded to the nearest integer?

A. 54 B. 52 C. 53 D. 55

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A. 661 B. 659 C. 662 D. 660

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A. 144 B. 145 C. 143 D. 146

Q14. Practice variant 2: Find the compound interest on Rs 1000 at 15% per annum for 2 years, compounded annually.

A. 323 B. 322 C. 321 D. 324

Q15. Practice variant 2: Rs 420 is divided in the ratio 3:3. What is the first share?

A. 211 B. 209 C. 210 D. 212

Q16. Practice variant 2: A invests Rs 1000 for 12 months and B invests Rs 1500 for 9 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?

A. 472 B. 470 C. 473 D. 471

Graded set - Questions 17-32

Q17. Practice variant 3: What is 25% of 240?

A. 60 B. 61 C. 59 D. 62

Q18. Practice variant 3: A value of 1000 rises by 12% and then falls by 7%. What is the final value?

A. 1043 B. 1042 C. 1041 D. 1044

Q19. Practice variant 3: Groups of 22 and 32 have averages 42 and 62. What is their combined average, rounded to the nearest integer?

A. 55 B. 53 C. 54 D. 56

Q20. Practice variant 3: An article costs Rs 600 and is sold at 20% profit. What is the selling price?

A. 721 B. 719 C. 722 D. 720

Q21. Practice variant 3: Find the simple interest on Rs 1400 at 7% per annum for 2 years.

A. 196 B. 197 C. 195 D. 198

Q22. Practice variant 3: Find the compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually.

A. 211 B. 210 C. 209 D. 212

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A. 275 B. 273 C. 274 D. 276

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A. 445 B. 443 C. 446 D. 444

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Q26. Practice variant 4: A value of 1000 rises by 13% and then falls by 8%. What is the final value?

A. 1041 B. 1040 C. 1039 D. 1042

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A. 256 B. 257 C. 255 D. 258

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A. 323 B. 322 C. 321 D. 324

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A. 217 B. 215 C. 216 D. 218

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A. 501 B. 499 C. 502 D. 500

Remedial and timed set - Questions 33-48

Q33. Practice variant 5: What is 20% of 280?

A. 56 B. 57 C. 55 D. 58

Q34. Practice variant 5: A value of 1000 rises by 14% and then falls by 9%. What is the final value?

A. 1038 B. 1037 C. 1036 D. 1039

Q35. Practice variant 5: Groups of 24 and 34 have averages 44 and 64. What is their combined average, rounded to the nearest integer?

A. 57 B. 55 C. 56 D. 58

Q36. Practice variant 5: An article costs Rs 700 and is sold at 20% profit. What is the selling price?

A. 841 B. 839 C. 842 D. 840

Q37. Practice variant 5: Find the simple interest on Rs 1800 at 9% per annum for 2 years.

A. 324 B. 325 C. 323 D. 326

Q38. Practice variant 5: Find the compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually.

A. 211 B. 210 C. 209 D. 212

Q39. Practice variant 5: Rs 600 is divided in the ratio 3:3. What is the first share?

A. 301 B. 299 C. 300 D. 302

Q40. Practice variant 5: A invests Rs 1000 for 12 months and B invests Rs 1500 for 9 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?

A. 472 B. 470 C. 473 D. 471

Q41. Practice variant 6: What is 25% of 300?

A. 75 B. 76 C. 74 D. 77

Q42. Practice variant 6: A value of 1000 rises by 15% and then falls by 10%. What is the final value?

A. 1036 B. 1035 C. 1034 D. 1037

Q43. Practice variant 6: Groups of 25 and 35 have averages 45 and 65. What is their combined average, rounded to the nearest integer?

A. 58 B. 56 C. 57 D. 59

Q44. Practice variant 6: An article costs Rs 750 and is sold at 20% profit. What is the selling price?

A. 901 B. 899 C. 902 D. 900

Q45. Practice variant 6: Find the simple interest on Rs 2000 at 10% per annum for 2 years.

A. 400 B. 401 C. 399 D. 402

Q46. Practice variant 6: Find the compound interest on Rs 1000 at 15% per annum for 2 years, compounded annually.

A. 323 B. 322 C. 321 D. 324

Q47. Practice variant 6: Rs 660 is divided in the ratio 4:3. What is the first share?

A. 378 B. 376 C. 377 D. 379

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A. 445 B. 443 C. 446 D. 444

PYQS AND ANSWER PRACTICE

Complete step-by-step solutions

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A. 30 B. 31 C. 29 D. 32

Correct answer: A. Percent means per hundred: $\text{base} \times \text{rate}/100$.

Q2. Practice variant 1: A value of 1000 rises by 10% and then falls by 5%. What is the final value?

A. 1046 B. 1045 C. 1044 D. 1047

Correct answer: B. Successive changes multiply growth factors; they do not simply cancel.

Q3. Practice variant 1: Groups of 20 and 30 have averages 40 and 60. What is their combined average, rounded to the nearest integer?

A. 53 B. 51 C. 52 D. 54

Correct answer: C. Combine totals, then divide by the combined count.

Q4. Practice variant 1: An article costs Rs 500 and is sold at 20% profit. What is the selling price?

A. 601 B. 599 C. 602 D. 600

Correct answer: D. Profit percent uses cost price as the base.

Q5. Practice variant 1: Find the simple interest on Rs 1000 at 5% per annum for 2 years.

A. 100 B. 101 C. 99 D. 102

Correct answer: A. $SI = PRT/100$, derived by adding the same annual interest on the original principal.

Q6. Practice variant 1: Find the compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually.

A. 211 B. 210 C. 209 D. 212

Correct answer: B. $\text{Amount} = P(1+r/100)^n$; subtract principal for compound interest.

Q7. Practice variant 1: Rs 360 is divided in the ratio 2:3. What is the first share?

A. 145 B. 143 C. 144 D. 146

Correct answer: C. One share equals total divided by total ratio-parts.

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A. 501 B. 499 C. 502 D. 500

Correct answer: D. Partnership shares are proportional to capital multiplied by time.

Q9. Practice variant 2: What is 20% of 220?

A. 44 B. 45 C. 43 D. 46

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A. 1036 B. 1035 C. 1034 D. 1037

Correct answer: B. Successive changes multiply growth factors; they do not simply cancel.

Q43. Practice variant 6: Groups of 25 and 35 have averages 45 and 65. What is their combined average, rounded to the nearest integer?

A. 58 B. 56 C. 57 D. 59

Correct answer: C. Combine totals, then divide by the combined count.

Q44. Practice variant 6: An article costs Rs 750 and is sold at 20% profit. What is the selling price?

A. 901 B. 899 C. 902 D. 900

Correct answer: D. Profit percent uses cost price as the base.

Q45. Practice variant 6: Find the simple interest on Rs 2000 at 10% per annum for 2 years.

A. 400 B. 401 C. 399 D. 402

Correct answer: A. $SI = PRT/100$, derived by adding the same annual interest on the original principal.

Q46. Practice variant 6: Find the compound interest on Rs 1000 at 15% per annum for 2 years, compounded annually.

A. 323 B. 322 C. 321 D. 324

Correct answer: B. Amount = $P(1+r/100)^n$; subtract principal for compound interest.

Q47. Practice variant 6: Rs 660 is divided in the ratio 4:3. What is the first share?

A. 378 B. 376 C. 377 D. 379

Correct answer: C. One share equals total divided by total ratio-parts.

Q48. Practice variant 6: A invests Rs 1000 for 12 months and B invests Rs 1500 for 10 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?

A. 445 B. 443 C. 446 D. 444

Correct answer: D. Partnership shares are proportional to capital multiplied by time.

Verified PYQ routing ledger

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2024	16	Mixture gain	A (supplied)	Conserve the amount of the active component.
2024	25	Partnership profit share	A (supplied)	Compare capital multiplied by time.
2024	26	Successive-percentage comparison	B (supplied)	Fix the base and multiply successive change factors.
2024	27	Inverse-operation error	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	28	Overlapping pass percentages	B (supplied)	Fix the base and multiply successive change factors.
2024	29	Age ratio	A (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	36	Proportional weights	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	47	Chained price ratios	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	10	Successive-percentage cycle	A (supplied)	Fix the base and multiply successive change factors.
2025	17	Fuel-cost distance adjustment	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	78	Average-balance relation	A (supplied)	Rebuild the total before and after the change.
2026	25	Class-average correction	B (provisional)	Rebuild the total before and after the change.
2026	30	Direct/inverse proportion	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	43	Percentage savings change	B (provisional)	Fix the base and multiply successive change factors.
2026	58	Company-worker ratio	B (provisional)	Add rates and invert the net rate.

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2026	65	Alloy-mixture ratio	A (provisional)	Conserve the amount of the active component.
2026	78	Partnership capital-duration share	A (provisional)	Compare capital multiplied by time.
2026	79	Repeated-replacement mixture	B (provisional)	Conserve the amount of the active component.

Final error-log checklist

- Was the family classified correctly?
- Were units, domains, quantifiers and constraints copied correctly?
- Was the first-principles relation written before a shortcut?
- Was the selected option independently recomputed or supported by the passage?
- Was the mistake logged and followed by a fresh remedial item?